

Comparative Financial Analysis of Dunkin' and Starbucks (2016-2019)

Bhoomika Chopra, Olivia Bennett, Weihao Lei, Shiqi Min, Tho Le, Kevin Cui

Brandeis University

Spring 2024

BUS-6A-2: Financial Accounting

Prof. Divya Khaitan

April 19th, 2024

Name	Work Contribution	Percentage
Bhoomika Chopra	Excel: - Starbucks Comparative Annual Balance Sheets - Starbucks Common-Size Balance Sheets - Starbucks Extended Dupont Analysis - Starbucks/Dunkins Benchmarks Report: - Ratio Analysis: Extended Dupont Analysis 10k Analysis (f)	16.66%
Olivia Bennett	Excel: - Starbucks Comparative Annual Income Statements - Starbucks Common-Size Income Statements - Starbucks Cash Flows Statements Report: 10k Analysis (e) - Investment Opportunity	16.66%
Weihaio Lei	Excel: - Starbucks Annual Stockholders Equity Statement - Starbucks Ratios based on DuPont Analysis Report: - Wrote an outline of 10-K Analysis (a-f) 10k Analysis (a-b)	16.66%
Shiqi Min	Excel: - Dunkins Comparative Annual Income Statements - Dunkins Common-Size Income Statements - Dunkins Statement of Cash Flows Report: - Ratio Analysis: Additional Ratio Analysis	16.66%
Tho Le	Excel: - Dunkins Annual Stockholders Equity Statement - Dunkins Ratios based on DuPont Analysis Report: - Introduction 10k Analysis (c-d)	16.66%
Kevin Cui	Excel: - Dunkins Comparative Annual Balance Sheets - Dunkins Common-Size Annual Balance Sheets - Dunkins Extended DuPont Analysis Report: - Common-Size Income Statements & Balance Sheets Review	16.66%

Introduction:

According to research's data published by the National Coffee Association, Coffee consumption in America has reached one of the highest peaks in history. In such a high-demand market, Starbucks and Dunkin' Donuts thrive as two of the biggest coffee chains in the world.

Starting their business in Seattle, 1971, Starbucks has become the one of the biggest coffee chains in America and the world. A decade later, Howard Schultz joined the company and brought his knowledge of Italian coffeehouse culture back to Starbucks and upscale its success. Starbucks' products include mainly many kinds of beverages revolving around coffees. They state in their mission statement: "We nurture the limitless possibilities of human connection."

Dunkin' Donuts was founded by William Rosenberg in Massachusetts in 1948. Dunkin' is famous for their coffees that are served along with other breakfast/food items such as donuts. Even though the chain is now owned by Inspire Brands, they still thrive with their mission to serve the finest coffees to their customers.

Being on top of their games, both chains are considered very popular. However, they would also have to compete with the local coffee shops and donut shops in the area where they have their franchise as well. Additionally, rising costs of coffee beans is a challenge facing the entire industry. In this analysis, both chains' financial activities and status will be studied.

Common-Size Income Statements & Balance Sheets Review:

For our Common-Size Financial statement review, Net Sales and Total Assets are used as base items for income statement and balance sheet, respectively. Starting with the Common-Size Income Statements for Dunkin from 2016-2019. Between 2016-2019, Total Operating Expenses decreased 3%, increasing the Operating Income by 2.5%. These changes could be attributed to the rise of mobile ordering, which automates the customer experience. Provision for income taxes decreased 7.3% in 2016-2017, then increased 3.5% between 2017-2018. The decrease in provision for income

taxes is due to FASB issuing “new guidance...which simplifies several aspects, including excess tax benefits”. The initial Tax Act passed in 2017 immediately reduced income taxes, but wore off in 2018 explaining the increase.

For the Common-Size Income Statements for Starbucks, the first significant change was for CFG, food services and other revenues in 2018 that declined 3.2% which was caused by selling their Tazo Brand, the licensing of CFG products to Nestle, and the closure of their E-commerce business in late 2017. Occupancy Expenses increased 9% in 2018, while Cost of Sales decreased 9%, due to Occupancy costs changing to be included with Occupancy Expenses, instead of with COGS. In 2018, net income increased by 6% due to a “gain from acquisition of a joint venture”.

Now we review Dunkin’s common-size balance sheets. Cash and cash equivalents spiked up over 10% in 2017, funded by long-term debt that they incurred. Net PPE in 2019 increased 9% due to it including Operating Lease Assets, unlike previous years. At the end of 2017, several Class A-2 notes increased long-term debt by 3%. Deferred Revenues increased 9% in 2018, due to new guidance in which franchise fees, advertising, and other transactions are included with Deferred revenues. Deferred Income Tax decreased by 6% in 2017, due to a “remeasurement of deferred tax liabilities using the lower enacted corporate tax rate”. The Additional Paid in Capital decreased by 7% in 2017, due to Dunkin repurchasing shares of common stock from shareholders and retiring shares of treasury stock. In 2019, Dunkin repurchased and retired stock again, which caused the 4% decline in APIC in 2019. In 2017 and 2019, net income was used to reduce accumulated deficit, which is the cause for the 10% and 10% decrease in accumulated deficit in the respective years. In 2018, Dunkin retired more treasury stock, increasing the accumulated deficit by 20%.

For Starbucks’ common-size balance sheet, cash and cash equivalents in 2018 increased by 20%. This change in cash came from SB's record net earnings and cash provided by changes in operating assets and liabilities. In 2019, cash and cash equivalents fell by 23% following a

repurchase of common stock. In 2018, Prepaid expenses and other current assets increased 3% due to increase in Income Tax receivable. In 2019, the income tax receivable dropped, halving Prepaid expenses and other current assets. In 2017, long-term investments dropped 4%, due to divesting in their Teavana and Tazo brands. Starbucks planned to continue these goals through 2018, which is why long-term investment dropped by another 2%. In 2018, deferred income taxes dropped 5%, and then rose 9% in 2019. Goodwill rose 4% in 2018, and rose again by 4% in 2019. The 2018 goodwill increase is attributed to the acquisition of the Chinese/Pacific Starbucks operations, while the change in 2019 is due to the decrease of Total Assets. Accrued liabilities fell 6% in 2018 as a result of Accrued payroll and benefits being accounted for separately for that year. In 2017 and 2018, Starbucks incurred several long-term notes, causing the rise of 5%, 10%, 20% respectively in long-term liabilities for years 2018 and 2019. Retained earnings fell over 30% in both 2018 and 2019 due to repurchasing of common stock.

Ratio Analysis:

I. Extended Dupont Analysis

a) The Asset Turnover Ratio ($\text{Revenue} / \text{Average Total Assets}$) assesses how effectively Starbucks and Dunkin' utilize their total assets to generate revenue, either by attracting more customers or maintaining customer retention rates. The Financial Leverage Ratio ($\text{Average Total Assets} / \text{Average Shareholders' Equity}$) reveals the proportion of Starbucks and Dunkin' total assets financed by shareholders' equity. Return on Assets or ROA ($\text{Net Income} / \text{Average Total Assets}$) evaluates the profitability of Starbucks and Dunkin' by comparing their revenue from customers minus operational costs (production expenses, inventory costs, and employee benefits) against their total assets. Return on Equity (ROE) gauges how efficiently Starbucks and Dunkin' utilize funds provided by owners or shareholders to generate profit, considering income from customers minus incurred costs.

b) From 2018 to 2019, there were multiple changes regarding the Dupont Analysis for Dunkin' and Starbucks. Dunkin's Financial Leverage ratio increased from -10 to -5.6. This points to large swings in Dunkin's company profits. This also played a factor in Dunkin's ROE increasing from -65% to -35%. Starbucks had many more drastic changes regarding the Dupont model compared to Dunkin'. The company's asset turnover ratio decreased from 1.73 to 1.22. Their financial leverage also decreased from 4.3 to -8.58. ROA for Starbucks also had a decrease from 0.32 to 0.17 in 2019. Starbucks ROE between 2018-2019 dropped from 136% to -142.40%.

c) These changes in Starbucks could be explained through the financial history of the company in 2018-19. Despite a 7% increase in total net revenues, operational challenges such as partner investments and strategic initiatives led to a compression in operating margin from 15.7% in 2018 to 15.4% in 2019, subsequently impacting the overall Profit Margin, which declined from 18.28% to 13.58%. Moreover, the Financial Leverage ratio shifted from 4.30 in 2018 to -8.58 in 2019 influenced by increased capital expenditures and higher shareholder returns. While capital expenditures decreased from \$2.0 billion in fiscal 2018 to \$1.8 billion in fiscal 2019, Starbucks returned \$12.0 billion to shareholders through share repurchases and dividends, compared to \$8.9 billion in fiscal 2018. Consequently, ROA decreased from 31.68% in 2018 to 16.60% in 2019, reflecting reduced profitability relative to total assets. Similarly, ROE experienced a decline from 136.24% in 2018 to -142.40% in 2019, indicating a drastic reduction in profitability relative to shareholder equity which can be attributed to the combined impact of decreased asset turnover, profit margin, and increased financial leverage, as outlined in the fiscal 2019 annual financial report. The changes in Dunkin's DuPont model can also be attributed to its financial performance in fiscal year 2019. The overall growth in systemwide sales of 4.6% was fueled by Dunkin' U.S. systemwide sales growth of 5.0%, resulting from new restaurant openings and comparable store sales growth driven by an increase in average ticket. Dunkin' International also contributed with a

7.6% systemwide sales growth, led by growth in various regions. Additionally, total revenues increased by \$48.6 million, or 3.7%, primarily driven by increases in franchise fees and royalty income, rental income, advertising fees, and other revenues. Overall, the changes in Dunkin's DuPont model reflect the company's efforts to drive sales growth, navigate currency fluctuations, and optimize revenue streams.

d) *Asset Turnover*: Starbucks: In 2019, Starbucks' Asset Turnover ratio decreased to 1.22 from 1.73 in 2018, while Dunkin's ratio increased slightly from 0.36 to 0.37. Both companies' ratios are significantly higher than the industry benchmark, indicating better utilization of assets to generate sales. *Profit Margin*: Starbucks: The profit margin decreased from 18.28% in 2018 to 13.58% in 2019, while Dunkin's increased from 17.40% to 17.66%. Both companies outperform the industry benchmark, showcasing higher profitability per dollar of sales. *Financial Leverage*: Starbucks: Experienced a drastic decrease in financial leverage from 4.30 in 2018 to -8.58 in 2019, indicating a shift towards a less leveraged capital structure. Dunkin' also decreased its leverage from -10.50 to -5.67. Both companies' leverage ratios are lower than the industry benchmark, suggesting a less risky financial position. *ROA (Return on Assets)*: Starbucks: ROA decreased from 31.68% in 2018 to 16.60% in 2019, whereas Dunkin's ROA increased from 6.22% to 6.56%. Both companies' ROA outperformed the industry benchmark, indicating efficient use of assets to generate profits. *ROE (Return on Equity)*: Starbucks: Experienced a drop in ROE from 136.24% in 2018 to -142.40% in 2019, reflecting a substantial decline in profitability relative to shareholders' equity. Dunkin's ROE increased from -65.28% to -37.21%. Starbucks ROE contrasts sharply with the industry benchmark, suggesting challenges in generating returns for shareholders compared to industry peers.

Overall, while both Starbucks and Dunkin' show strengths in certain financial metrics compared to industry benchmarks, such as higher asset turnover, profit margins, and ROA, Starbucks declining ROEs indicate challenges in generating returns for shareholders compared to industry

standards. Dunkin's increase in ROE shows growth in generating returns to shareholders.

Additionally, Starbucks' significant drop in financial leverage in 2019 suggests a notable shift in its capital structure.

II. Additional Ratios Analysis

a) The Quality of Income ratio evaluates Dunkin' or Starbucks' cash flow generation from core business activities, particularly focusing on their sales of key products.

The Cash ratio assesses Dunkin' and Starbucks' liquidity holding and ability to meet short-term liabilities using cash and cash equivalents, without relying on other assets.

Debt to Equity measures the proportion of liability versus equity in Dunkin' and Starbucks. Higher debt to equity means more borrowing, and indicates higher risk level.

Dividend yield measures Dunkin' and Starbucks' dividend payments to shareholders. A lower yield may indicate reinvestment of earnings for potential future growth.

The Price/Earnings ratio compares Dunkin' and Starbucks' stock prices to their Earnings per share. It offers a more accurate view of market perception than EPS, which can be manipulated.

b) This part highlights changes in Dunkin' and Starbucks from 2018 to 2019. According to Table 1, Dunkin' saw a 31% decrease in Debt to Equity. According to Table 2, Starbucks' quality of income fell by 47% to 1.4, cash ratio decreased by 71% to 0.44, and Debt to Equity ratio decreased dramatically by 121% to -4.08. Conversely, Starbucks' P/E ratio increased by 55% to 30.75.

c) Dunkin's significant decrease in Debt to Equity stems from rising liabilities and stockholders' equity. Total liabilities increased by \$339,000 thousands, primarily driven by higher operating and capital lease liabilities, along with accounts payable. This signals business expansion or financial stress if Dunkin' struggles to repay leases. Meanwhile, stockholders' equity rose by \$125,000 thousands due to a reduction in Accumulated deficit. Despite this, Dunkin's total

Stockholders equity was still negative, leading to a decrease in Debt to Equity ratio. Furthermore, drastic stock repurchases in 2017 and 2018 totaling \$807,000 thousands are the main reason for negative equity, potentially as preparation for privatization.

Starbucks' decrease in its Debt to Equity is due to a massive drop in stockholders' equity, decreasing to deficit. The total stockholders' equity fell from \$1.17 billion to -\$6.23 billion, due to a major repurchase of common stock totaling \$10.13 billion. Concurrently, Starbucks' total liabilities increased by \$2.47 billion in 2019, driven by rises in income tax payable and long-term debt. These shifts in equity and liabilities led to a decline in Debt to Equity. The stock buybacks indicate confidence or an effort to boost share prices and EPS, explaining the increase in the P/E ratio from 64 to 90. However, Starbucks also experienced a decrease in cash ratio due to a sudden drop in cash & cash equivalents, influenced by declining net income, and increased outflows in various areas, especially financing activities due to stock repurchases. Posing potential challenges for Starbucks in repaying its liabilities.

d) Both saw decreased Quality of Income, with Dunkin' at 1.23 and Starbucks at 1.4. Despite this, Starbucks performed slightly better in cash flow generation. Dunkin's Cash ratio rose to 1.07 during the year, indicating higher liquidity and easier repayment of current liabilities. In contrast, Starbucks' Cash ratio dropped significantly to 0.44 in 2019, down from 1.54 in 2018. Both companies' Debt to Equity ratios decreased over the year, Dunkin' decreased from -5.85 to -7.67, and Starbucks dramatically decreased from 19.65 to -4.08. Both companies experienced stockholders deficit, which is mainly due to repurchases of common stock. Corresponding to the decreasing ROE of both companies, this suggests challenges in finance. While Starbucks experienced a more drastic decrease, implying greater challenge. Both companies' Dividends yield ratio decreased throughout the year. Dunkin decreased to 2.02%, and Starbucks to 0.45%. This suggests higher market price per share, which corresponds to the huge stock repurchase of both

companies. Dunkin's Dividends yield is higher than Starbucks, which may be more attractive to investors. Both companies' P/E ratio increased, Dunkin's P/E increased to 25.42, while Starbucks to 30.75. Starbucks having higher P/E suggests higher expectation from the market, which leads to higher market share price. Through comparison, Starbucks slightly outperformed in cash flow generation. Dunkin's improved Cash ratio indicates better liquidity. While both companies' Debt to Equity ratios decreased, Starbucks decreased more, hurting its ROE. Decreased Dividend Yield ratios reflect higher market prices driven by stock repurchases in both companies. Both also saw increased P/E ratios, signaling higher market expectations, particularly in Starbucks, suggesting higher share prices.

10-K Analysis:

a) Starbucks Corporation uses the average cost method to value its inventory. This method helps to smooth out price fluctuations in the cost of inventory and makes the cost of goods sold and ending inventory value less sensitive to changes in the price of individual items. This method is helpful for a company like Starbucks, which deals with a large number of raw materials, such as coffee beans, whose prices can vary depending on market conditions. For Dunkin' Donuts, inventory is also valued using the average cost method. Like Starbucks, Dunkin' benefits from this method because it mitigates the effects of fluctuating prices of basic ingredients such as coffee and dairy products. By using the average cost method, Dunkin' can maintain consistency in product pricing, which is essential to retain customers in fast-food industries.

b) Depreciation of Starbucks Corporation's property, plant and equipment is computed using the straight-line method. This method is applied over the estimated useful lives of the assets, which generally range from 2 to 15 years for equipment and 30 to 40 years for buildings. Dunkin' Donuts uses the straight-line method of depreciation for property and equipment, including software. This method allocates the cost of an asset equally over its expected useful life.

c) In the years of 2017 and 2018, Starbucks signed an agreement to purchase the remaining 50% ownership of East China joint venture from Uni-President Enterprises Corporation (“UPEC”) and President Chain Store Corporation (“PCSC”) in order to acquire their controls and operations of the chain in mainland China. This agreement puts Starbucks in an estimated total liability of \$1.3 billion dollars which is what the 50% ownership is worth.

Dunkin’ Donuts’ contingent liability is based on agreements and contracts. Dunkin’ Donuts have had, up to 2019, many agreements with suppliers for their franchisees. A lot of these agreements are one year or less. However, if Dunkin’ fails to obligate their agreements with the suppliers, they can face a potential liability of \$100.9 millions dollars. If this number happens and is recorded on the balance sheet, their liability will spike and the company’s liquidity ratio will be lowered.

d) In 2018, Dunkin’ Donuts made a repurchase for common stocks worth \$680 millions after they retired a large amount of their stocks. Compared to their repurchases of common stock made in the next year of almost \$30 million, their purchase for treasury stocks in 2019 is significant and will also reduce their outstanding stocks. The similar big amount of repurchased stocks is also observed for Starbucks in 2018 which was \$7 billions. Though in 2019, they continue to do so and this time, they repurchased \$10 billions worth of common stocks. For both of the companies, the sales of stocks were not significant.

e) In 2019, Starbucks’ highest cash inflow from operating activities was from depreciation and amortization due to high depreciation expense. The highest inflow from operating that actually contributed to the cash amount was the increase of income tax payables. The highest cash outflow from operating activities was deferred income taxes. From investing activities, the biggest cash outflow was additions to PPE. From financing activities, the highest cash outflow was from the repurchase of stock, and the highest inflow is from proceeds from issuance of debt. The major

sources of cash for Starbucks are from the proceeds of issuance of debt, and the increase of income tax payables. The major uses of cash from Starbucks are the additions to PPE and the repurchase of stock. For Dunkin, the highest inflow of cash in operating activities is from depreciation and amortization, due to high depreciation expense, and the highest outflow is from net income of equity method investments. From investing activities, the highest outflow was the purchase of PPE. In financing activities, the highest outflow from repayment of long-term debt and the highest inflow from the issuance of long-term debt. The major sources of cash for Dunkin are from the issuance of long-term debt, while the major uses of cash are the purchase of PPE and the repayment of long-term debt.

f) Starbucks's Independent Audit report was audited by Deloitte & Touche LLP and Dunkin's Independent Audit report was audited by KPMG LLP. Both audits expressed an "unqualified opinion" on the financial statements of both companies. According to reputable independent audit firms, both reports affirm that Starbucks and Dunkin currently adhere to the Generally Accepted Accounting Principles (GAAP) and are considered fair.

Investment Opportunity:

If there was 500,000 to invest in Dunkin or Starbucks I would choose Dunkin. Both companies have consistent operating incomes, although Dunkin's operating expense has decreased in 2019 due to increased mobile ordering. This shows Dunkin is utilizing technology to reduce expenses which is a good sign for investors. Additionally, Dunkin's ROE increased whereas Starbucks' ROE decreased, showing a reduction in profitability relative to shareholder equity. The P/E ratio measures a company's market price relative to its EPS, the lower the better. Starbucks has a P/E ratio of 30.75 while Dunkin has a P/E ratio of 25.42, showing that investing in Dunkin would have you earn more relative to the market price. Having a Debt to Equity ratio below 0.1 is a good sign. Starbucks has a Debt to Equity ratio of -4.08 and Dunkin has one of -7.67. The smaller the

Debt to Equity ratio is, the more the company is using its assets rather than borrowing money, which means Dunkin is utilizing its assets more than Starbucks. Although Dunkin's ratios show it is a better investment, Starbucks has returned more money back to shareholders through dividends and share repurchases than Dunkin. Starbucks returned 12 billion whereas Dunkin returned 36 million. That being said, due to better ratio showings, I would invest in Dunkin.

Bibliography:

All Industries: Industry financial ratios benchmarking. (n.d.). Ready Ratios. Retrieved from

<https://www.readyratios.com/sec/industry/>

Debt to Equity Ratio: Meaning, Types, Benefits, & Limitations (n.d), NirmilBang. Retrieved from

<https://www.nirmalbang.com/knowledge-center/debt-to-equity-ratio.html#:~:text=A%20high%20debt%2Dto%2Dequity%20ratio%20indicates%20that%20a%20company,debt%2Dto%2Dequity%20ratio.>

Dunkin' Brands. (2017). Annual Report 2017. PDF

Dunkin' Brands. (2018). Annual Report 2018. PDF

Dunkin' Brands. (2019). Annual Report 2019. PDF

Dunkin' Brands' (DNKN) Stock Down Despite Q4 Earnings Beat. (2020, February 7). Nasdaq.

Retrieved from

<https://www.nasdaq.com/articles/dunkin-brands-dnkn-stock-down-despite-q4-earnings-beat-2020-02-07>

Dunkin' Brands Group, INC. Annual Report Pursuant to Section 13 or 15(d) of the Securities

Exchange Act OF 1934. Retrieved from

<https://www.sec.gov/Archives/edgar/data/1357204/000135720420000015/dnkn-20191228x10k.htm>

Starbucks. (2017). Fiscal 2017 Annual Report. Retrieved from

https://s22.q4cdn.com/869488222/files/doc_financials/annual/2017/01/FY17-Starbucks-Form-10-K.pdf

Starbucks. (2018). Fiscal 2018 Annual Report. Retrieved from

https://s22.q4cdn.com/869488222/files/doc_financials/annual/2018/2018-Annual-Report.p

Starbucks. (2019). Fiscal 2019 Annual Report. Retrieved from

https://s22.q4cdn.com/869488222/files/doc_financials/2019/2019-Annual-Report.pdf

Starbucks Corporation, Starbucks Reports Q4 and Full Year Fiscal 2019 Results. (n.d.). Starbucks

Investor Relations. Retrieved from

<https://investor.starbucks.com/press-releases/financial-releases/press-release-details/2019/Starbucks-Reports-Q4-and-Full-Year-Fiscal-2019-Results/default.aspx>

The Economic Impact of the Coffee Industry. (n.d.). NCA. Retrieved from

<https://www.ncausa.org/Research-Trends/Economic-Impact>

Tincher Laurel. (n.d.). How to Analyze Stocks: 4 Ways. SoFi. Retrieved from

<https://www.sofi.com/learn/content/how-to-analyze-a-stock/#:~:text=One%20of%20the%20most%20common,ratio%20is%20a%20good%20sign.>

Appendix Exhibits:

Ratios	Formula	2019	2018
Quality of Income	Cash flow from operating activities / Net income	1.23	1.17
Cash Ratio	Cash + cash equivalents / Current liability	1.07	0.96
Debt to Equity	Total liabilities / Stockholders equity	-7.67	-5.85
Dividends yield	Dividends per share / Market price per share	2.02%	2.19%
Price/Earnings	Market share price / EPS	25.42	23.85

Table 1: Additional ratios of Dunkin' from year 2018 to 2019

Ratios	Formula	2019	2018
Quality of Income	Cash flow from operating activities / Net income	1.4	2.63
Cash Ratio	Cash + cash equivalents / Current liability	0.44	1.54
Debt to Equity	Total liabilities / Stockholders equity	-4.08	19.65
Dividends yield	Dividends per share / Market price per share	0.45%	0.71%
Price/Earnings	Market share price / EPS	30.75	19.88

Table 2: Additional ratios of Starbucks from year 2018 to 2019